

DEEPANA BALMOOR

Backend Engineer | Java • Spring Boot • Kafka • PostgreSQL • Microservices

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SUMMARY

Backend engineer with 2+ years designing REST APIs, Kafka/JMS messaging and custom Java logic for a high-volume telecom platform in production, now focused on code-first Java and Spring Boot development. Built a Kafka-based order system in Spring Boot with a persisted Saga orchestrator, transactional outbox, idempotent consumers and Testcontainers-tested failure handling. Strong in failure handling, production debugging (Elastic, Grafana) and go-live support across five batches covering 55M+ subscribers.

TECHNICAL SKILLS

Languages & Frameworks: Java, Spring Boot, Spring MVC, Spring Data JPA/Hibernate, SQL, Python, C++ | JUnit, Mockito, Postman

APIs & Architecture: REST API design, Swagger/OpenAPI, microservices, Saga pattern, idempotency, event-driven design

Data & Messaging: PostgreSQL, MySQL, Oracle, MongoDB, Redis, Apache Kafka, JMS/EMS, TIBCO ActiveSpaces

DevOps & Observability: Docker, Kubernetes, OpenShift, CI/CD, Linux, Git/GitHub, Maven, Grafana, Elastic

Integration Middleware: TIBCO BWCE/Flogo, Mashery API gateway, SOAP/REST integration

Problem Solving: Leetcode - 411 problems solved | AI-assisted development: Claude Code, Copilot

PROJECTS

Distributed Order Management System

Java 17, Spring Boot 3, Kafka, PostgreSQL, Flyway, Testcontainers, Docker

github.com/dbalmoor/distributed-order-system

- Built Order, Inventory and Payment services with a persisted Saga orchestrator over Kafka, each service owning its own PostgreSQL database (Flyway migrations); compensation handles payment failure, timeouts and cancellation, with payment success as the point of no return.
- Implemented a transactional outbox (lease-based poller with **FOR UPDATE SKIP LOCKED**), idempotent consumers and guards against late or out-of-order events, for at-least-once delivery without duplicate side effects.
- Added a saga watchdog with per-step deadlines, backoff retries and a **NEEDS_ATTENTION** state with operator retry/force-resolve APIs; Kafka retry topics, dead-letter topics and an offset-based DLT replay endpoint.
- Wrote Testcontainers integration tests (Kafka, PostgreSQL) for failure scenarios; optimistic locking and MDC correlation IDs for tracing across services.

Spring Boot POCs

Java, Spring Boot, Spring Kafka

- Re-implemented the DMP API in Spring Boot with feature parity and deployed it; currently building a Spring Kafka integration POC.

PROFESSIONAL EXPERIENCE

Software Engineer, Tech Mahindra

Aug 2024 – Present

Client: Indosat Ooredoo Hutchison (IOH), Indonesia | Digital Monetization Platform (DMP) Consolidation | TIBCO BWCE/Flogo

- Delivered the project's first API on schedule (SIT/UAT passed) and raised development throughput by 30% in the first week; within a month was independently deploying to OpenShift and exposing APIs via Mashery.
- Wrote a design document from the existing code base, cutting teammates' investigation time by 50% and easing ops handover.
- Found JVM properties for BWCE deployments in performance testing: pod utilization up 60%, CPU cores down 40%.
- Wrote custom Java in BWCE applications for HTTPS integration, MD5 hashing, cache management (auto and manual refresh) and complex business logic; JavaScript in Flogo for complex logic.
- Integrated channels and downstream systems through BWCE: DVM(Digital Vending Machine), CVM (Offer Management), SMS, Flash Fulfillment and EOC (Customer and Product Orders).
- Used Kafka in nearly all callback APIs, alongside JMS/EMS flows, ActiveSpaces caching and PostgreSQL/Oracle sync; designed REST/SOAP services with retries and compensating transactions; Docker deployments on OpenShift.
- Signed off SIT and UAT one month ahead of schedule; debugged production issues using Elastic, Grafana and database state, and root-caused and fixed two operations-reported functional defects quickly, reducing complaints.
- Used GitHub Copilot and Claude Code for code review, test generation and understanding the existing codebase, reviewing and validating every suggestion before it went in
- Received the "Pat on the Back" award for delivering critical modules under tight timelines without hurting production stability.

International On-site Deployment, IOH Jakarta, Indonesia

Apr-May 2026, Aug-Sep 2026

- Supported all five go-live batches totaling 55M subscribers (2 offshore: 15M; 3 on-site: 40M).

- Ran pre-go-live “mini-market” validation (apps deployed to production and tested standalone without switching traffic); supported triage and defect analysis and closed regression at 100%.
- Supported ~60% of channel-wise night traffic switches; 98% of switches completed with no revenue loss.
- Triaged production through go-live; closed with a 30-day hypercare (half the planned time).

EDUCATION

B.Tech, Information Technology (CGPA 8.94), Geethanjali College of Engineering and Technology

2020 – 2024

- Tech Mahindra training: 6 months (Java, Python, SQL, 95% score) + 1 month BWCE/Flogo project training.